

# Computing Skills and Python

**Student Number: 760072671**

**Module Code: HPDM206Z**

**Assignment Title: Github Coding Task and Short Report**

**Date: 30/09/2026**

## Introduction

The purpose of this document is to provide evidence of learning in the stated module and showcase my work. The SQL (Structured Query Language) queries below have been used in an SQL database named 'hospital\_db' and data extracted from the hospital.csv, doctors.csv, patients.csv, and prescriptions.csv. I have created four tables named respectively of the corresponding data in the csv files.

## SQL Queries

### Doctors at a Hospital Query

**Purpose:** To print a list of doctors at a particular hospital.

#### **Query:**

```
SELECT Doctors.Name, Hospital.Name
FROM Doctors
INNER JOIN Hospital
ON Doctors.HospitalID = Hospital.HospitalID
WHERE Hospital.Name = 'Miller and Sons Hospital';
```

#### **Output:**

|                           |                          |
|---------------------------|--------------------------|
| Dr. Carl Jones            | Miller and Sons Hospital |
| Dr. Jennifer Peterson DVM | Miller and Sons Hospital |

|                       |                          |
|-----------------------|--------------------------|
| Dr. Jacob Hardy       | Miller and Sons Hospital |
| Dr. Edward Harrington | Miller and Sons Hospital |
| Dr. Alison Adams      | Miller and Sons Hospital |
| Dr. Mark Craig        | Miller and Sons Hospital |

### Patient's Prescriptions Query

**Purpose:** To print a list of prescription for a particular patient, ordered by the prescription date.

#### **Query:**

```
SELECT Prescriptions.DrugName, Prescriptions.DateIssued
FROM Prescriptions
INNER JOIN Patients
ON Prescriptions.PatientID = Patients.PatientID
WHERE Patients.PatientID = '323'
ORDER BY Prescriptions.DateIssued ASC;
```

#### **Output:**

| DrugName    | DateIssued |
|-------------|------------|
| +-----+     | +-----+    |
| Losartan    | 2023-08-02 |
| Amoxicillin | 2023-12-22 |
| Ibuprofen   | 2024-04-30 |
| +-----+     | +-----+    |

3 rows in set (0.00 sec)

### Doctor's Prescriptions Query

**Purpose:** Print a list of all prescriptions that a particular doctor has prescribed.

**Query:**

```
SELECT Prescriptions.DrugName, Doctors.Name
FROM Prescriptions
INNER JOIN Doctors
ON Prescriptions.DoctorID = Doctors.DoctorID
WHERE Doctors.Name = 'Dr. Devin Cox';
```

**Output:**

| DrugName    | Name          |
|-------------|---------------|
| Simvastatin | Dr. Devin Cox |
| Simvastatin | Dr. Devin Cox |
| Metformin   | Dr. Devin Cox |

3 rows in set (0.00 sec)

Add a Patient Query

**Purpose:** Add a new patient to the database, including being registered with one of the doctors.

**Query:**

```
INSERT INTO Patients (PatientID, Name, DOB, DoctorID) VALUES (701, 'Gerald Constance', '1975-05-20', 98);
```

```
SELECT * From Patients WHERE PatientID = 701;
```

**Output:**

| PatientID | Name | DOB | DoctorID |
|-----------|------|-----|----------|
|-----------|------|-----|----------|

|                           |     |                  |            |  |    |  |
|---------------------------|-----|------------------|------------|--|----|--|
|                           | 701 | Gerald Constance | 1975-05-20 |  | 98 |  |
| +-----+-----+-----+-----+ |     |                  |            |  |    |  |

1 row in set (0.00 sec)

### Most Prescribing Doctor Query

**Purpose:** Identify which doctor has made the most prescriptions.

**Query:**

```
SELECT Doctors.DoctorID, Doctors.Name, COUNT(*) AS TotalPrescriptions
FROM Prescriptions
INNER JOIN Doctors
ON Prescriptions.DoctorID = Doctors.DoctorID
GROUP BY Doctors.DoctorID
ORDER BY TotalPrescriptions DESC
LIMIT 1;
```

**Output:**

|                           |          |                    |  |                    |  |
|---------------------------|----------|--------------------|--|--------------------|--|
|                           | DoctorID | Name               |  | TotalPrescriptions |  |
| +-----+-----+-----+-----+ |          |                    |  |                    |  |
|                           | 19       | Dr. Taylor Sanchez |  | 13                 |  |
| +-----+-----+-----+-----+ |          |                    |  |                    |  |

1 row in set (0.00 sec)

### Doctors at Largest Hospital Query

**Purpose:** Print a list of all doctors at the hospital with the biggest size.

**Query:**

```
SELECT Doctors.Name
```

```

FROM Doctors
INNER JOIN Hospital
ON Doctors.HospitalID = Hospital.HospitalID
WHERE Hospital.HospitalID = (
SELECT Hospital.HospitalID
FROM Hospital
ORDER BY Hospital.Beds DESC
LIMIT 1
);

```

**Output:**

```

| Name                |
+-----+
| Dr. Kathryn Ray     |
| Dr. Regina Ryan     |
| Dr. Carly Moyer     |
| Dr. Scott Reynolds  |
+-----+
4 rows in set (0.01 sec)

```